[C, None, 9.2] For the circuit in Figure 0.1, assume a unit delay through the Register and Logic blocks (i.e., $t_R = t_L = 1$). Assume that the registers, which are positive edge-triggered, have a set-up time t_S of 1. The delay through the multiplexer t_M equals 2 t_R.

a. Determine the minimum clock period. Disregard clock skew.

b. Repeat part a, factoring in a nonzero clock skew: $\delta = t'_\theta - t_\theta = 1$.

c. Repeat part a, factoring in a non-zero clock skew: $\delta = t'_\theta - t_\theta = 4$.

d. Derive the maximum positive clock skew that can be tolerated before the circuit fails.

e. Derive the maximum negative clock skew that can be tolerated before the circuit fails.

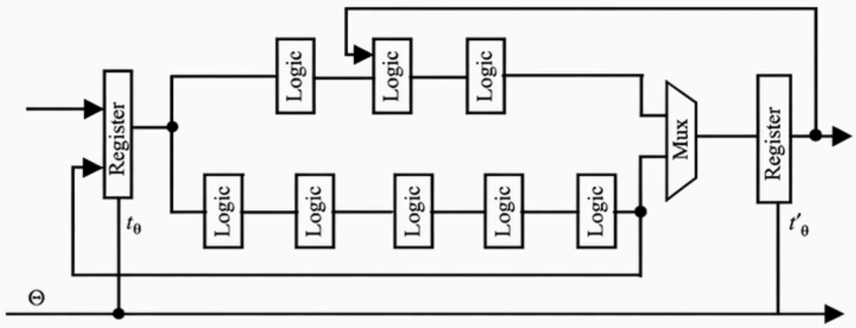

Figure 0.1 Sequential circuit.

a. Determine the minimum clock period. Disregard clock skew.

Solution

The circuit and paths of interest has been reproduced for convenience in Figure 0.1

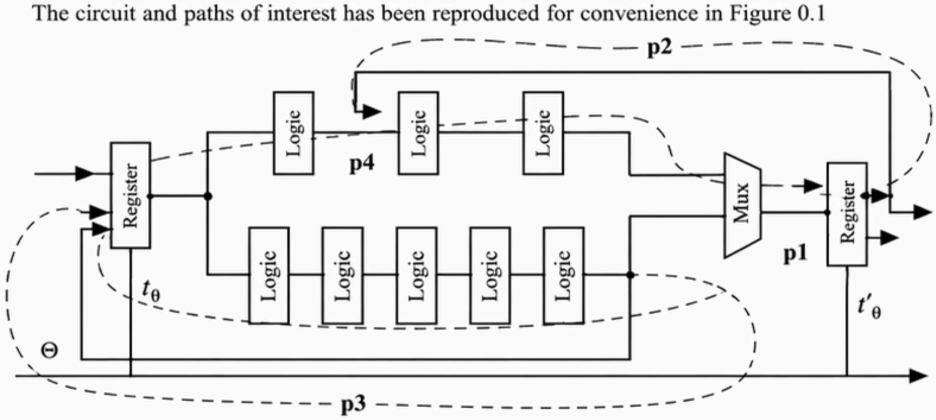

Figure 0.1 Sequential circuit.

Out of the 4 paths shown in the figure, **p1** is the critical one and determines the lower bound on the clock period. Using $T \geq t_{reg} + t_{logic} + t_{setup} - \delta$, we get $\boldsymbol{T_{min}} = \mathbf{1 + 7 + 1 = 9}$.

b. Repeat part a, factoring in a nonzero clock skew: $\delta = t'_\theta - t_\theta = 1$.

Solution

With finite clock skew, the time periods for different paths are as follows : $T_{min}(\text{p1}) = 9 - 1 = 8$, $T_{min}(\text{p2}) = 6$, $T_{min}(\text{p3}) = 7$, $T_{min}(\text{p4}) = 7 - \mathbf{1} = 6$ (Note that the clock skew is 0 for paths p2 and p3). Therefore the minimum clock period is $\boldsymbol{T_{min}} = \mathbf{8}$.

c. Repeat part a, factoring in a non-zero clock skew: $\delta = t'_\theta - t_\theta = 4$.

Solution

As the clock skew increases, the most significant path changes. Repeating the calculations in part (b) we get : $T_{min}(\text{p1}) = 9 - 4 = 5$, $T_{min}(\text{p2}) = 6$, $T_{min}(\text{p3}) = 7$, $T_{min}(\text{p4}) = 7 - 4 = 3$ (Note that the clock skew is 0 for paths p2 and p3). Therefore the minimum clock period is $\boldsymbol{T_{min}} = 7$.

d. Derive the maximum positive clock skew that can be tolerated before the circuit fails.

Solution

The maximum positive clock skew is determined by the inequality $\delta \leq t_{cd,reg} + t_{cd,logic}$. Assuming that the contamination delay is same as the propagation delay, we get $\boldsymbol{\delta_{max}} = \mathbf{1 + 3 + 2 = 6}$. Note that p4 determines the maximum tolerable skew (the fastest path will produce the earliest contamination). Paths p3 and p2 do not matter since there is no skew involved.

e. Derive the maximum negative clock skew that can be tolerated before the circuit fails.

Solution

The maximum positive negative skew has no bound since the clock period has no upper bound.